

Azure Cloud Data Lakehouse — System Architecture

Legend:

- Azure Services
- Databricks
- ADLS / Delta Lake
- Synapse
- Great Expectations
- Airflow
- Terraform
- GitHub Actions

SOURCES

Orders API
REST · 6-hourly

Clickstream API
REST · Hourly

Inventory SFTP
CSV · Daily

Customers API
REST · Daily

INGESTION

Azure Data Factory
5 Pipelines · Master Orchestration

ADLS Gen2

- Bronze
- Silver
- Gold

PROCESSING

Azure Databricks (PySpark)

Bronze → Silver

- 3 Hubs (customer, product, order)
- 2 Links (order-customer, order-product)
- 4 Satellites (SCD2 via hash_diff)

Silver → Gold

- gold_orders_wide (denormalized)
- gold_customer_360 (lifetime)
- gold_inventory_snapshot
- gold_daily_sales_agg

Delta Lake

- ACID Transactions
- Time Travel
- Z-Order Indexing
- Auto-Optimize
- VACUUM + COMPACT

SERVING + OPS

Azure Synapse Analytics
4 External Tables · 5 Views

Great Expectations
3 Suites · Quality Gate Plugin

Apache Airflow
2 DAGs · Orchestration

Terraform
4 Modules IaC

Docker Compose
Local Dev Environment

GitHub Actions
pytest · flake8 · CI/CD